Area in Polar Coordinates

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The polar area formula**

State the formula for the area enclosed by a polar curve $r = f(\theta)$ between $\theta = \alpha$ and $\theta = \beta$.

$A = \frac{1}{2}\int_{\alpha}^{\beta} r^2\,d\theta = \frac{1}{2}\int_{\alpha}^{\beta} [f(\theta)]^2\,d\theta.$

2 **[Foundational]** • **Target: The polar area formula**

Find the area enclosed by one full loop of $r = 3\cos\theta$ (i.e. $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$).

$A = \frac{1}{2}\int_{-\pi/2}^{\pi/2} 9\cos^2\theta\,d\theta = \frac{9}{2}\int_{-\pi/2}^{\pi/2} \frac{1+\cos 2\theta}{2}\,d\theta = \frac{9}{4}\left[\theta + \frac{\sin 2\theta}{2}\right]_{-\pi/2}^{\pi/2} = \frac{9}{4}(\pi) = \frac{9\pi}{4}.$

3 **[Foundational]** • **Target: Area between two polar curves**

Find the area inside the circle $r = 3$ but outside the circle $r = 2$.

$A = \frac{1}{2}\int_{0}^{2\pi} (3^2 - 2^2)d\theta = \frac{1}{2}\int_{0}^{2\pi} 5\,d\theta = \frac{1}{2}(5)(2\pi) = 5\pi.$

4 **[Foundational]** • **Target: Area of one petal of a rose curve**

Find the area of one petal of $r = 2\sin(2\theta)$, using the interval $\theta \in [0, \frac{\pi}{2}]$ where one petal is traced.

$A = \frac{1}{2}\int_{0}^{\pi/2} 4\sin^2(2\theta)\,d\theta = 2\int_{0}^{\pi/2} \frac{1-\cos 4\theta}{2}\,d\theta = \left[\theta - \frac{\sin 4\theta}{4}\right]_{0}^{\pi/2} = \frac{\pi}{2}.$

5 **[Foundational]** • **Target: Setting up area between curves that intersect**

Set up (but do not evaluate) the integral for the area inside both $r = 2$ and $r = 2 + 2\cos\theta$, noting they intersect where $2 = 2 + 2\cos\theta$, i.e. $\theta = \pm\frac{\pi}{2}$.

By symmetry, for $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ the circle $r = 2$ is the inner (smaller) boundary, while for $\theta \in [\frac{\pi}{2}, \frac{3\pi}{2}]$ the cardioid is inner. Area $= \frac{1}{2}\int_{-\pi/2}^{\pi/2} 2^2\,d\theta + \frac{1}{2}\int_{\pi/2}^{3\pi/2} (2+2\cos\theta)^2\,d\theta.$

6 **[Foundational]** • **Target: Area of a full rose curve**

Find the total area enclosed by all 4 petals of $r = 3\cos(2\theta)$, using the fact that one petal is traced for $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$.

Area of one petal: $\frac{1}{2}\int_{-\pi/4}^{\pi/4} 9\cos^2(2\theta)\,d\theta = \frac{9}{2}\int_{-\pi/4}^{\pi/4} \frac{1+\cos 4\theta}{2}\,d\theta = \frac{9}{4}\left[\theta + \frac{\sin 4\theta}{4}\right]_{-\pi/4}^{\pi/4} = \frac{9}{4}\cdot\frac{\pi}{2} = \frac{9\pi}{8}$. Total for all 4 petals: $4 \times \frac{9\pi}{8} = \frac{9\pi}{2}$.

7 **[Foundational]** • **Target: Area inside a limaçon**

Find the area enclosed by the limaçon $r = 3 + \cos\theta$ (a curve without an inner loop, since the constant exceeds the coefficient).

$A = \frac{1}{2}\int_{0}^{2\pi} (3+\cos\theta)^2\,d\theta = \frac{1}{2}\int_{0}^{2\pi} (9 + 6\cos\theta + \cos^2\theta)d\theta = \frac{1}{2}\left[9\theta + 6\sin\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_{0}^{2\pi} = \frac{1}{2}(18\pi + \pi) = \frac{19\pi}{2}.$

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Area inside a limaçon**

Find the area of the small inner loop of the limaçon $r = 1 + 2\cos\theta$ (constant less than coefficient, so an inner loop exists), traced for $\theta \in \left[\frac{2\pi}{3}, \frac{4\pi}{3}\right]$.

$A = \frac{1}{2}\int_{2\pi/3}^{4\pi/3}(1 + 2\cos\theta)^2\,d\theta = \frac{1}{2}\int_{2\pi/3}^{4\pi/3}(1 + 4\cos\theta + 4\cos^2\theta)d\theta = \frac{1}{2}[3\theta + 4\sin\theta + \sin 2\theta]_{2\pi/3}^{4\pi/3}$. At $\frac{4\pi}{3}$: $4\pi - 2\sqrt{3} + \frac{\sqrt{3}}{2}$. At $\frac{2\pi}{3}$: $2\pi + 2\sqrt{3} - \frac{\sqrt{3}}{2}$.
Difference: $2\pi - 3\sqrt{3}$. $A = \frac{1}{2}\left(2\pi - 3\sqrt{3}\right) = \pi - \frac{3\sqrt{3}}{2} \approx 0.54$.

9 **[Intermediate]** • **Target: Setting up area problems (AP-style, no evaluation)**

Set up (but do not evaluate) the integral for the area inside the rose $r = 2\sin(3\theta)$ that lies in the first quadrant only, for one petal traced on $\theta \in [0, \frac{\pi}{3}]$.

$A = \frac{1}{2}\int_0^{\pi/3}[2\sin(3\theta)]^2\,d\theta = \frac{1}{2}\int_0^{\pi/3}4\sin^2(3\theta)\,d\theta$.

10 **[Intermediate]** • **Target: Setting up area problems (AP-style, no evaluation)**

Set up (but do not evaluate) the integral for the area between the circle $r = 5$ and the cardioid $r = 5 - 5\sin\theta$ (inside the circle, outside the cardioid), for $\theta \in [0, \pi]$.

Area $= \frac{1}{2}\int_0^{\pi}[5^2 - (5 - 5\sin\theta)^2]d\theta$.

11 **[Intermediate]** • **Target: Finding intersection points of polar curves**

Find all points of intersection of $r = 2\cos\theta$ and $r = 2\sin\theta$ (both circles of radius 1) by setting the equations equal.

$2\cos\theta = 2\sin\theta \Rightarrow \tan\theta = 1 \Rightarrow \theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$. At $\theta = \frac{\pi}{4}$: $r = 2\cos\left(\frac{\pi}{4}\right) = \sqrt{2}$. At $\theta = \frac{5\pi}{4}$: $r = 2\cos\left(\frac{5\pi}{4}\right) = -\sqrt{2}$ (a negative r, representing the same point reached differently). Both curves also pass through the pole ($r = 0$) at different angles, which must be checked separately.

12 **[Intermediate]** • **Target: Finding intersection points of polar curves**

Explain why checking intersections of two polar curves algebraically (setting $f(\theta) = g(\theta)$) can MISS the pole as an intersection point, and why it must be checked separately.

Different curves can reach the pole ($r = 0$) at different values of θ (since a single point, the origin, corresponds to $r = 0$ for ANY θ). Setting $f(\theta) = g(\theta)$ only finds angles where both curves have the SAME r at the SAME θ, so it can miss cases where each curve passes through the origin at a different angle — the pole must always be checked as a separate case.

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13 **[Intermediate]** • **Target: Area using symmetry to simplify computation**

Explain how the symmetry of $r = 3\cos(2\theta)$ about both axes allows the total area to be found by computing just one petal and multiplying, rather than integrating over the full range at once.

Since $r = 3\cos(2\theta)$ produces 4 congruent petals by symmetry (reflections across both the x- and y-axes map petals onto each other), computing the area of ONE petal via a single definite integral and multiplying by 4 avoids having to integrate piecewise across the full $[0, 2\pi]$ range while tracking sign changes.

14 **[Intermediate]** • **Target: Area review: full worked example**

Find the area enclosed by $r = \sqrt{\theta}$ for $\theta \in [0, \pi]$ (a spiral arc closed off by the segment back to the pole).

$A = \frac{1}{2}\int_0^{\pi} (\sqrt{\theta})^2 \, d\theta = \frac{1}{2}\int_0^{\pi} \theta \, d\theta = \frac{1}{2}\left[\frac{\theta^2}{2}\right]_0^{\pi} = \frac{\pi^2}{4}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Area review: full worked example** • **Cross-Lesson: 1.01 Integration by Parts**

Find the area enclosed by $r = 4\sin\theta$ (a circle) for $\theta \in [0, \pi]$, and verify your answer matches the known area formula for a circle of the appropriate radius.

$A = \frac{1}{2}\int_0^{\pi} 16\sin^2\theta \, d\theta = 8\int_0^{\pi} \frac{1-\cos 2\theta}{2} \, d\theta = 4\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi} = 4\pi$. This matches πr^2 for a circle of radius 2 (since $r = 4\sin\theta$ traces a circle of radius 2 centered at $(0, 2)$): $\pi(2)^2 = 4\pi$ ✓.

16 **[Challenging]** • **Target: Comparing polar area to rectangular area (conceptual)** • **Cross-Lesson: 1.02 Partial Fractions and Improper Integrals**

Explain why the polar area formula $A = \frac{1}{2}\int r^2 \, d\theta$ uses r^2 rather than just r, by relating it to the area of a thin circular sector.

A thin sector of angle $d\theta$ and radius r has area approximately $\frac{1}{2}r^2 \, d\theta$ (a fraction $\frac{d\theta}{2\pi}$ of the full circle's area πr^2). Summing (integrating) these thin sector areas over the range of θ gives the total polar area formula — the r^2 arises naturally from the geometry of a circular sector, not from any arbitrary convention.

17 **[Challenging]** • **Target: Comparing polar area to rectangular area (conceptual)** • **Cross-Lesson: 2.01 Euler's Method and Logistic Growth**

A student claims the area between $r = 1$ and $r = 2$ (an annulus, full circle) is simply $\frac{1}{2}\int_0^{2\pi} (2-1)^2 \, d\theta$. Explain the error and give the correct setup.

The error is subtracting the radii before squaring — the correct polar formula requires subtracting the SQUARED radii, not squaring the difference of radii. Correct setup:
$A = \frac{1}{2}\int_0^{2\pi} (2^2 - 1^2) d\theta = \frac{1}{2}\int_0^{2\pi} 3 \, d\theta = 3\pi$ (matching $\pi(2^2 - 1^2) = 3\pi$ from the standard annulus-area formula).

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18 [Challenging] • Target: Area of a full rose curve • Cross-Lesson: 3.01 Parametric Equations

Level 1 · FoundationsConvert the Cartesian equation $x^2 + y^2 = 6y$ into polar form, and identify the resulting geometric shape.

Substitute $x^2 + y^2 = r^2$ and $y = r\sin\theta$: $r^2 = 6r\sin\theta \Rightarrow \mathbf{r = 6\sin\theta}$. In Cartesian coordinates, $x^2 + (y^2 - 6y + 9) = 9 \Rightarrow x^2 + (y-3)^2 = 3^2$, which is a circle with radius 3 centered at $(0, 3)$.

19 [Challenging] • Target: Area inside a limaçon • Cross-Lesson: 3.02 Vector-Valued Functions

Use a two-term Maclaurin polynomial for e^x to approximate $e^{0.1}$.

Method · Taylor/Maclaurin Polynomial Approximation: Step 1 · Define the two-term (degree 1) Maclaurin polynomial for e^x: $P_1(x) = 1 + x$. Step 2 · Evaluate at $x = 0.1$: $P_1(0.1) = 1 + 0.1 = \mathbf{1.1}$. Step 3 · Error analysis via Lagrange Remainder: By Taylor's Theorem: $|R_1(0.1)| = \left|\frac{f''(c)}{2!}(0.1)^2\right| = \frac{e^c}{2}(0.01)$ for some $c \in (0, 0.1)$. Since $e^c < e^{0.1} < 1.2$, $|R_1(0.1)| < \frac{1.2}{2}(0.01) = 0.006$. Exact value: $e^{0.1} \approx 1.10517$. The tw.

20 [Challenging] • Target: Setting up area problems (AP-style, no evaluation) • Cross-Lesson: 4.01 Polar Curves and Derivatives

Level 1 · FoundationsEvaluate the improper integral $\int_1^{\infty} \frac{1}{x^3}\,dx$ using proper limit notation, or state that it diverges.

Rewrite as a limit: $\lim_{b\to\infty} \int_1^b x^{-3}\,dx = \lim_{b\to\infty} \left[-\frac{1}{2x^2}\right]_1^b = \lim_{b\to\infty} \left(-\frac{1}{2b^2} - \left(-\frac{1}{2(1)^2}\right)\right)$. Since $\lim_{b\to\infty} \frac{1}{2b^2} = 0$: $0 - \left(-\frac{1}{2}\right) = \mathbf{\frac{1}{2}}$. The integral converges to $\frac{1}{2}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The polar area formula

Find the area enclosed by $r = 1$ for $\theta \in [0, \frac{\pi}{2}]$.

a. $\frac{\pi}{4}$ ✓

b. $\frac{\pi}{2}$

c. $-\frac{7\pi}{4}$

d. $\frac{3\pi}{4}$

$A = \frac{1}{2}\int_0^{\frac{\pi}{2}} 1^2\,d\theta = \frac{1}{2}(1)(\frac{\pi}{2}) = \frac{\pi}{4}$.

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22 [Foundational] • Target: Area of a full cardioid

Find the total area enclosed by the curve $r = 8 + 7\sin\theta$.

a. 113π

b. $\frac{177\pi}{2}$ ✓

c. $\frac{179\pi}{2}$

d. $\frac{225\pi}{2}$

Expanding $(8 + 7\sin\theta)^2 = 8^2 + 2(8)(7)\sin\theta + 7^2\sin^2\theta$ and integrating over $[0, 2\pi]$, the linear $\sin\theta$ term vanishes and $\sin^2\theta$ averages to $\frac{1}{2}$, leaving $A = \pi\left(8^2 + \frac{7^2}{2}\right) = \frac{177\pi}{2}$.

23 [Foundational] • Target: Area between two polar curves

Find the area of one petal of $r = 4\cos(2\theta)$.

a. $\frac{9\pi}{4}$

b. 2π ✓

c. 4π

d. 8π

Using $A_{petal} = \frac{k^2\pi}{4n}$ with $k = 4$, $n = 2$: $A = \frac{16\pi}{8} = 2\pi$.

24 [Foundational] • Target: Area of one petal of a rose curve

Find the area inside the circle $r = 9$ but outside the circle $r = 2$, for $\theta \in [0, \pi]$ only (half the full region).

a. 77π

b. $\frac{79\pi}{2}$

c. $\frac{77\pi}{2}$ ✓

d. $\frac{49\pi}{2}$

$A = \frac{1}{2}\int_0^{\pi}(9^2 - 2^2)d\theta = \frac{1}{2}(77)(\pi) = \frac{77\pi}{2}$.

25 [Foundational] • Target: Setting up area between curves that intersect

Consider the parametric curve defined by $x(t) = 3\cos(t) - \cos(3t)$ and $y(t) = 3\sin(t) - \sin(3t)$ for $t \in [0, 2\pi]$. This curve traces out a shape known as a 'cardioid'. Find the maximum area of a triangle that can be inscribed within the loop of this cardioid, with one vertex at the origin.

a. $\frac{27\sqrt{3}}{2}$

b. $\frac{81}{2}$

c. $\frac{81\sqrt{3}}{4}$ ✓

d. $54\sqrt{3}$

The parametric equations describe a cardioid. We need to find the loop of the cardioid. Let's analyze the behavior of $x(t)$ and $y(t)$. The curve forms a loop when the path self-intersects. This typically happens when the parameter values lead to the same point. To find the area of a triangle with one vertex at the origin and the other two vertices on the curve, we can use the formula for the area of a triangle given two sides and the included angle, or by using the determinant formula: Area =.

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26 [Foundational] • Target: Area of a full rose curve

Polar area between two curves $r_1(\theta) \le r_2(\theta)$:

a. $\frac{1}{2}\int (r_2^2 - r_1^2)d\theta$ ✓

b. $\frac{1}{2}\int (r_2 - r_1)^2 d\theta$

c. $\int (r_2 - r_1)d\theta$

d. $\pi(r_2^2 - r_1^2)$

$A = \frac{1}{2}\int (r_2^2 - r_1^2)d\theta$.

27 [Foundational] • Target: Area inside a limaçon

Consider a region in the first quadrant bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis. A rectangle is inscribed within this region such that one vertex is at the origin, two sides lie along the x and y axes, and the fourth vertex lies on the curve $y = 9 - x^2$. If the area of this inscribed rectangle is maximized, determine the value of the definite integral that represents the volume of a solid formed by revolving the region bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis about the y-axis.

a. $\frac{81\pi}{2}$ ✓

b. 81π

c. 162π

d. $\frac{243\pi}{2}$

Let the vertex of the inscribed rectangle on the curve be (x, y). Since the vertex is on $y = 9 - x^2$, and the rectangle is in the first quadrant, $x \ge 0$ and $y \ge 0$. The dimensions of the rectangle are x and y. The area of the rectangle is $A = xy$. Substituting $y = 9 - x^2$, we get $A(x) = x(9 - x^2) = 9x - x^3$ for $0 \le x \le 3$. To maximize the area, we find the derivative of $A(x)$ with respect to x and set it to zero: $A'(x) = 9 - 3x^2$ Setting $A'(x) = 0$ gives $9 - 3x^2 = 0$, so.

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28 [Foundational] • Target: Setting up area problems (AP-style, no evaluation)

A rectangular garden is to be enclosed by a fence. One side of the garden is adjacent to a long straight river, and requires no fencing. The other three sides require fencing, and the total length of the fence available is 120 meters. Let the side of the garden parallel to the river have length x meters, and let the two sides perpendicular to the river each have length y meters. The area of the garden is given by $A(x, y) = xy$. \\ Find the dimensions x and y that maximize the area of the garden.

a. **x = 60 m, y = 30 m** ✓

b. x = 30 m, y = 60 m

c. x = 40 m, y = 40 m

d. x = 80 m, y = 20 m

The total length of the fence is 120 meters. The fence is used for the three sides of the rectangle not along the river. Thus, the constraint equation is $x + 2y = 120$. \\ We want to maximize the area $A = xy$. From the constraint equation, we can express x in terms of y: $x = 120 - 2y$. \\ Substitute this expression for x into the area formula: $A(y) = (120 - 2y)y = 120y - 2y^2$. \\ To find the maximum area, we need to find the critical points of $A(y)$ by taking the derivative with respect to y and setti.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Finding intersection points of polar curves

Find the area of one petal of the rose $r = 1\cos(3\theta)$ (3 odd).

a. $\frac{\pi}{12}$ ✓

b. $\frac{\pi}{6}$

c. $\frac{\pi}{4}$

d. $\frac{\pi}{24}$

One petal spans $\theta \in [-\pi/6, \pi/6]$: $A = \frac{1}{2}\int r^2\, d\theta = \frac{\pi}{12}$.

30 [Intermediate] • Target: Area using symmetry to simplify computation

Area of one petal of $r = 2\sin 3\theta$:

a. $\pi/3$ ✓

b. $2\pi/3$

c. π

d. $\pi/6$

$A = \frac{1}{2}\int_0^{\pi/3} 4\sin^2(3\theta) d\theta = \pi/3$.

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31 [Intermediate] • Target: Area review: full worked example

Limaçon $r = a + b\sin\theta$. For $a = b = 1$ (cardioid), area:

a. $\pi/2$

b. $3\pi/2$ ✓

c. 2π

d. π

$A = \frac{1}{2}\int_0^{2\pi}(1+\sin\theta)^2 d\theta = 3\pi/2$.

32 [Intermediate] • Target: Comparing polar area to rectangular area (conceptual)

Polar area $A = \frac{1}{2}\int r^2\, d\theta$. For one petal of $r = \sin 2\theta$:

a. $\pi/4$

b. $\pi/8$ ✓

c. $\pi/2$

d. 1

$A = \frac{1}{2}\int_0^{\pi/2}\sin^2(2\theta) d\theta = \pi/8$.

33 [Intermediate] • Target: The polar area formula

The area of the region enclosed by $x = t - \sin t, y = 1 - \cos t$ (one arch of cycloid) using $A = \int y\, dx$:

a. πr^2

b. $3\pi r^2$ **(with $r = 1$)** ✓

c. π

d. 2π

One arch of cycloid: area $= 3\pi r^2$. With $r = 1$: 3π.

34 [Intermediate] • Target: Area of a full cardioid

Area swept inside one loop of $r^2 = 4\sin 2\theta$:

a. 1

b. 2 ✓

c. 4

d. π

$A = \frac{1}{2}\int_0^{\pi/2} 4\sin 2\theta\, d\theta = 2$.

35 [Intermediate] • Target: Area between two polar curves

Area of region bounded by parametric curve $x = \cos^3 t, y = \sin^3 t$ (full astroid):

a. $3\pi/8$ ✓

b. $3\pi/4$

c. $\pi/4$

d. $\pi/2$

Astroid area $= 3\pi a^2/8$ with $a = 1$: $3\pi/8$.

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36 [Intermediate] • Target: Area of one petal of a rose curve

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points where $f(x) = 0$. Determine the sum of the x-coordinates of these intersection points.

a. 1

b. 3

c. **4** ✓

d. -3

The x-intercepts of the graph of $y = f(x)$ are the roots of the equation $f(x) = 0$. For a quadratic equation of the form $ax^2 + bx + c = 0$, the sum of the roots is given by $-\frac{b}{a}$. In this case, $f(x) = x^2 - 4x + 3$, so $a = 1$, $b = -4$, and $c = 3$. The sum of the x-coordinates of the intersection points is $-\frac{-4}{1} = 4$.

37 [Intermediate] • Target: Setting up area between curves that intersect

Consider the function $f(x) = -x^2 + 4x - 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at two points. What is the value of the x-coordinate of the point of intersection with the larger x-value?

a. 1

b. 2

c. **3** ✓

d. 4

To find the x-intercepts, we set $f(x) = 0$. This gives the quadratic equation $-x^2 + 4x - 3 = 0$. Multiplying by -1, we get $x^2 - 4x + 3 = 0$. Factoring this equation, we have $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. The larger x-value is 3.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Area of a full rose curve • Cross-Lesson: 3.01 Parametric Equations

If $|f^{(4+1)}(c)| \le 1$ on the relevant interval, bound the Lagrange error $|R_4(x)|$ for approximating f at $x = 0.1$ using a Taylor polynomial centred at $a = 1$.

a. 0.0098415

b. **0.00492075** ✓

c. 0.00246038

d. 0.0273375

Lagrange error bound: $|R_4(x)| \le \frac{M|x-a|^{4+1}}{(4+1)!} = \frac{1 \cdot 0.9^5}{5!} \approx 0.00492075$.

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39 [Challenging] • Target: Area inside a limaçon • Cross-Lesson: 3.02 Vector-Valued Functions

Suppose $|f^{(3+1)}(c)| \le 3$ for all c between 1 and 0.1. Bound $|R_3(0.1)|$ for the Taylor polynomial centred at $a = 1$. (extra variant)

a. 0.0410063

b. **0.0820125** ✓

c. 0.3645

d. 0.164025

$|R_3(x)| \le \frac{M|x-a|^{3+1}}{(3+1)!} \approx 0.0820125.$

40 [Challenging] • Target: Setting up area problems (AP-style, no evaluation) • Cross-Lesson: 4.01 Polar Curves and Derivatives

Euler's method approximates solutions by using:

a. Second derivatives

b. **Tangent line steps** ✓

c. Exact antiderivatives

d. Taylor polynomials of degree 2

Euler's method: $y_{n+1} = y_n + h \cdot f(x_n, y_n)$

41 [Challenging] • Target: Finding intersection points of polar curves • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

For $\frac{dy}{dx} = 2y,\ y(0) = 1$, use one Euler step of size $h = 0.25$ to approximate $y(0.25)$.

a. **1.5** ✓

b. 0.5

c. 2

d. 1.25

$y_1 = y_0 + h\,f(x_0, y_0) = 1 + 0.25(2 \cdot 1) = 1.5.$

42 [Challenging] • Target: Area using symmetry to simplify computation • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

For $\frac{dy}{dx} = x + y,\ y(0) = 1$, take one Euler step of size $h = 0.5$ to approximate $y(0.5)$.

a. 1

b. **1.5** ✓

c. 2

d. 1.55

$y_1 = y_0 + h\,f(x_0, y_0) = 1 + 0.5(0 + 1) = 1.5.$

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43 [Challenging] • Target: Area review: full worked example • Cross-Lesson: 6.01 Power Series and Interval of Convergence

A rectangular field is to be enclosed by a fence. One side of the field is adjacent to a river and does not require fencing. The farmer has 120 meters of fencing material. Let the dimensions of the field be x meters perpendicular to the river and y meters parallel to the river. The area of the field is given by $A = xy$. Part A: Express the area A as a function of x. Part B: Determine the dimensions x and y that maximize the area of the field. Provide the maximum possible area.

a. $x = 20$ m, $y = 80$ m, $A_{\max} = 1600$ m^2

b. $x = 30$ m, $y = 60$ m, $A_{\max} = 1800$ m^2 ✓

c. $x = 40$ m, $y = 40$ m, $A_{\max} = 1600$ m^2

d. $x = 25$ m, $y = 70$ m, $A_{\max} = 1750$ m^2

Part A: The total length of fencing is $2x + y = 120$. Thus, $y = 120 - 2x$. Substituting this into the area formula, we get $A(x) = x(120 - 2x) = 120x - 2x^2$. Part B: To find the maximum area, we need to find the critical points of $A(x)$ by taking the derivative and setting it to zero. $A'(x) = \frac{dA}{dx} = 120 - 4x$. Setting $A'(x) = 0$: $120 - 4x = 0 \Rightarrow 4x = 120 \Rightarrow x = 30$ meters. To confirm this is a maximum, we can use the second derivative test: $A''(x) = \frac{d^2A}{dx^2} = -4$. Si.

44 [Challenging] • Target: Comparing polar area to rectangular area (conceptual) • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Find the area inside $r = 2\cos(\theta)$ and outside $r = 1$.

a. π

b. $\sqrt{3} + \frac{2\pi}{3}$

c. $\frac{\sqrt{3}}{2} + \frac{\pi}{3}$ ✓

d. $\frac{\pi}{3}$

The curves meet at $\theta = \pm\pi/3$. $A = \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \left(2\cos^2(\theta) - \frac{1}{2}\right) d\theta = \frac{\sqrt{3}}{2} + \frac{\pi}{3}$.

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45 [Challenging] • Target: The polar area formula • Cross-Lesson: 1.01 Integration by Parts

Which formula gives the length of a polar curve $r = f(\theta)$?

a. $\int_{\alpha}^{\beta} r\,d\theta$

b. $\int_{\alpha}^{\beta} |dr/d\theta|\,d\theta$

c. $\int_{\alpha}^{\beta} r^2\,d\theta$

d. $\int_{\alpha}^{\beta} \sqrt{r^2 + (dr/d\theta)^2}\,d\theta$ ✓

Applying Applications of Integration principles, Polar arc length formula. Therefore, the verified correct option is $\int_{\alpha}^{\beta} \sqrt{r^2 + (dr/d\theta)^2}\,d\theta$.

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